

DREAM FABRICATION DOCUMENT

LAYER STACKUP LEGEND

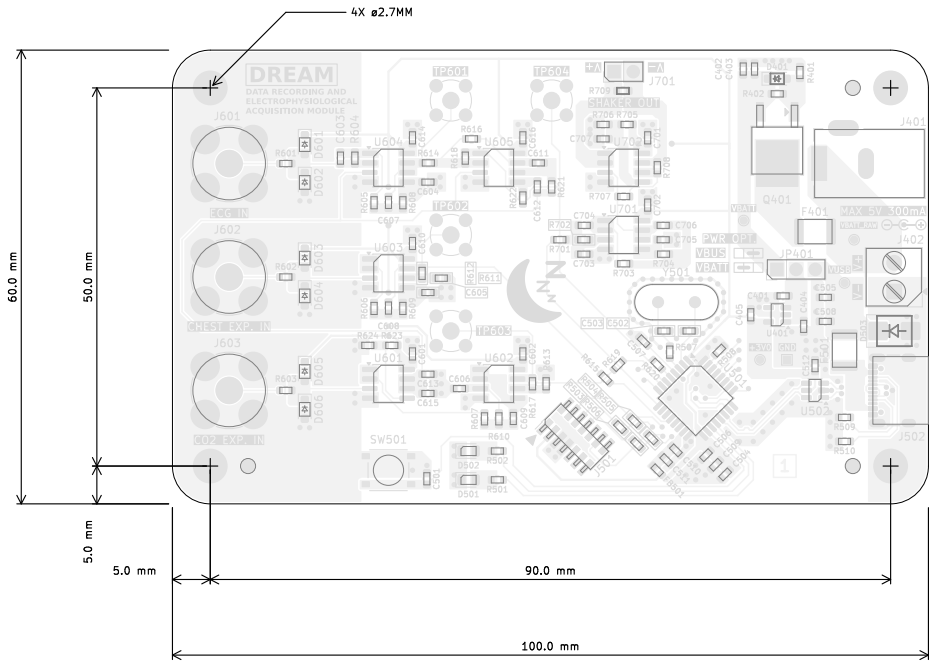
	Material	Layer	Thickness	Dielectric	Type	Gerber
	F,Paste				Paste Mask	
	F,Silkscreen				Legend	GBR
	F,Mask		0,02mm		Solder Resist	GBR
	Copper	L1 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
	Core		1,49mm	FR4_7628	Dielectric	
	Copper	L2 (Sig, GND)	0,035mm (1,00oz)		Plane	GBR
	B,Mask		0,02mm		Solder Resist	GBR
	B,Silkscreen				Legend	GBR
	B,Paste				Paste Mask	

Total thickness: 1,6mm
Note: external layer thicknesses are specified after plating

TOP FABRICATION (SCALE 1:1)

IMPEDANCE TABLE

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
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ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- FABRICATE PER IPC-6012A CLASS 2.
- OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.
- SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- SURFACE FINISH: IMMERSION GOLD
- SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- ALL VIAS ARE TO BE NON-TENTED.
- VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- PCB MATERIAL REQUIREMENTS:
 - FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - Tg 170 C OR EQUIVALENT.
 - EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY RYAN DYNAMICS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	100,000 × 60,000 mm
BOARD THICKNESS	1,600 mm
TRACE WIDTH	0,200 mm
TRACE TO TRACE	0,200 mm
MIN. HOLE (PTH)	0,300 mm
MIN. HOLE (NPTH)	0,650 mm
ANNULAR RING	0,150 mm
COPPER TO HOLE	0,254 mm
COPPER TO EDGE	0,250 mm
HOLE TO HOLE	0,254 mm

PROJECT
MENG420 PROJ

DRN
2026-04-12 R. HICKS
CHK

ENG APP

MFR APP

SHEET PATH

FILENAME
DREAM.kicad_pcb



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DREAM

GIT HASH

315b6dd

DRAWING No

FABRICATION

A3

VARIANT

RELEASED

REVISION

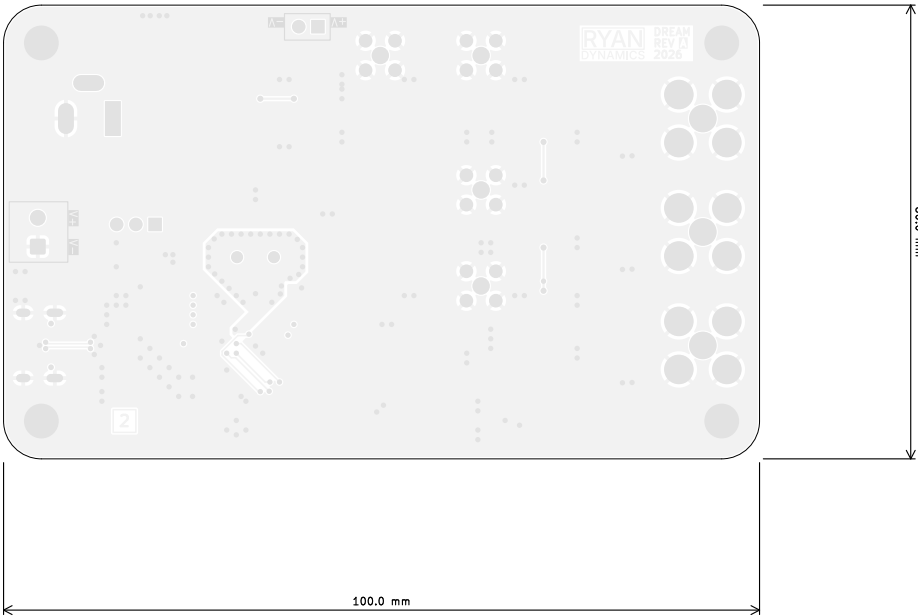
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SHEET

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DREAM FABRICATION DOCUMENT

BOTTOM FABRICATION (SCALE 1:1)



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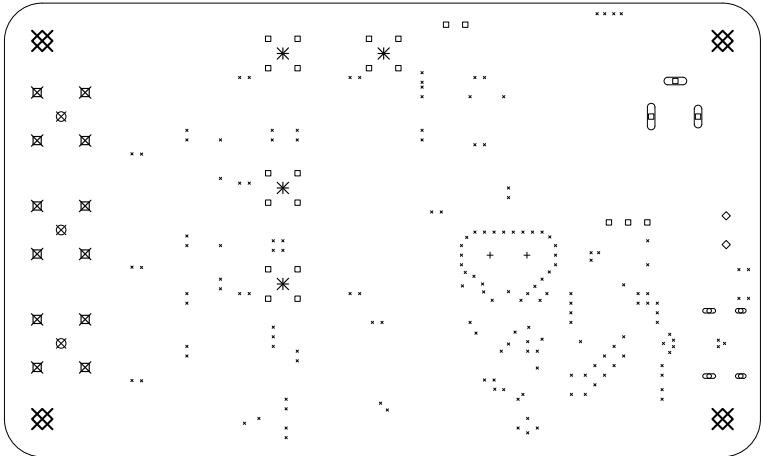
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DREAM FABRICATION DOCUMENT

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	169	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Via
○	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
+	2	0,80mm (31,50mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
□	21 + 3	1,00mm (39,37mils)	PTH	Round + Slot	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
◇	2	1,10mm (43,31mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊗	3	1,27mm (50,00mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
✱	4	1,50mm (59,06mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊗	12	1,52mm (60,00mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊗	4	2,70mm (106,30mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
Total 224						

DRILL DRAWING L1 – L2 (SCALE 1:1)



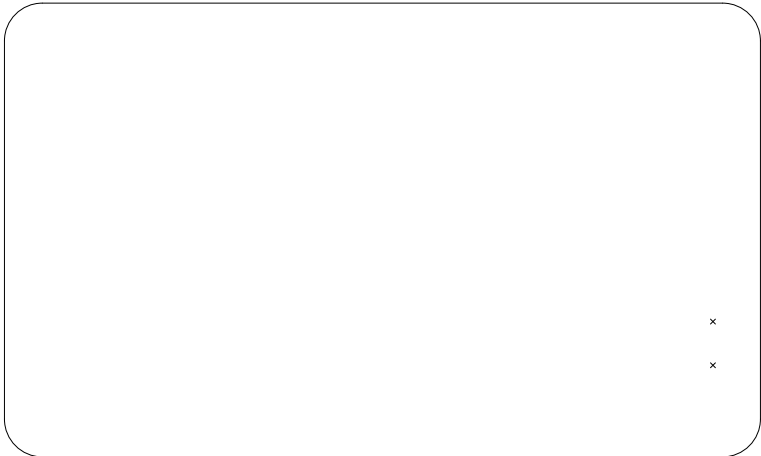
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SHEET PATH	GIT HASH 315b6dd	DRAWING No	FABRICATION A3
FILENAME DREAM.kicad_pcb	VARIANT RELEASED	REVISION 1.2.0	
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DREAM FABRICATION DOCUMENT

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	2	0.65mm (25.59mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Mechanical
	Total 2					

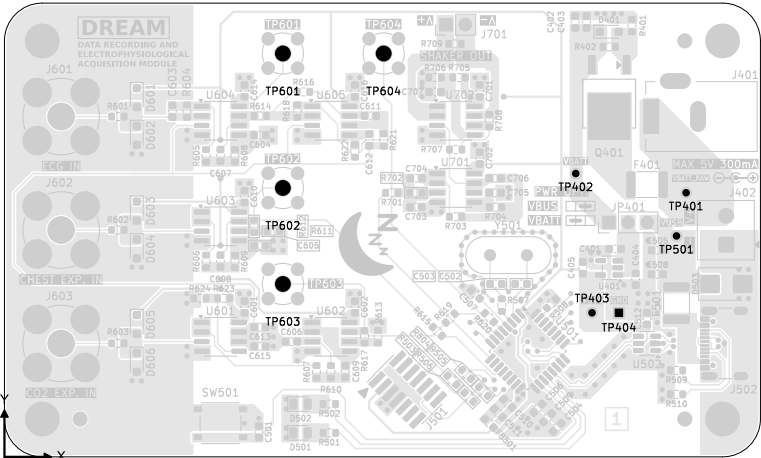
DRILL DRAWING L1 – L2 (SCALE 1:1)



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SHEET PATH	GIT HASH 315b6dd	DRAWING No	FABRICATION	
FILENAME DREAM.kicad_pcb	VARIANT RELEASED	REVISION 1.2.0		
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DREAM FABRICATION DOCUMENT

TOP TEST POINTS (SCALE 1:1)



Ref.	Net	X [mm]	Y [mm]
TP401	VBATT_RAW	90.17	34.92
TP402	VBATT	75.56	37.46
TP403	+3V0	77.72	19.05
TP404	GND	81.28	19.05
TP501	VUSB	88.90	29.21
TP601	ECG_FILTER	36.83	53.34
TP602	CE_FILTER_GND	36.83	35.56
TP603	CO2_OUT	36.83	22.86
TP604	ECG_OUT_GND	50.16	53.34

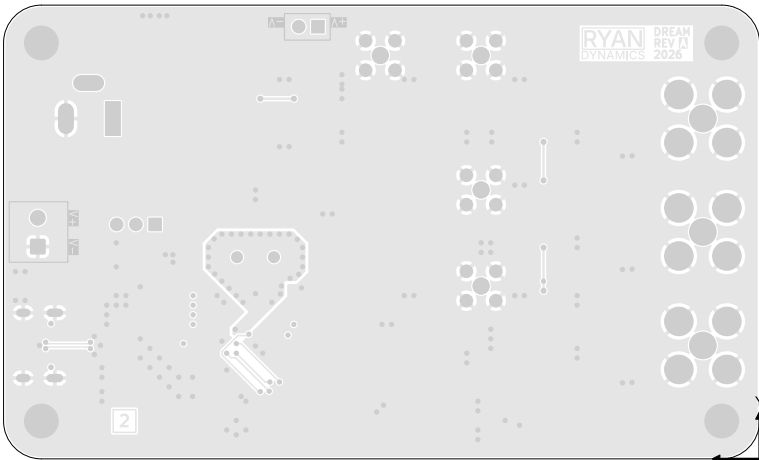
Ref.	Net	X [mm]	Y [mm]
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FILENAME	DREAM.kicad_pcb	VARIANT	RELEASED
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DREAM FABRICATION DOCUMENT

BOTTOM TEST POINTS (SCALE 1:1)



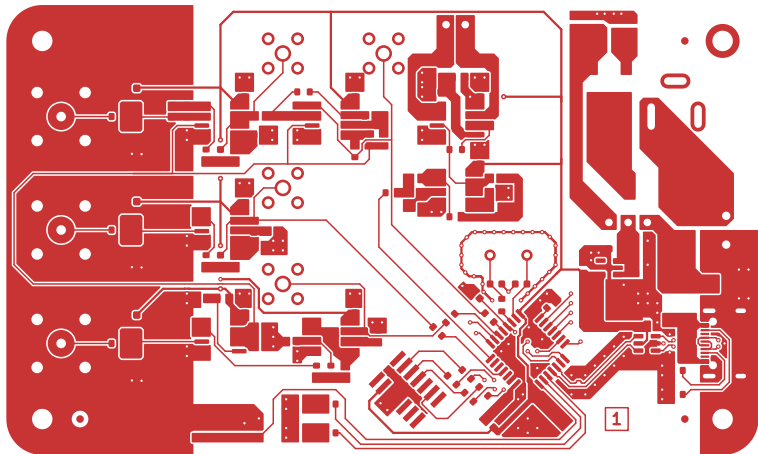
Ref.	Net	X [mm]	Y [mm]
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DREAM FABRICATION DOCUMENT

L1 (Sig, PWR) (SCALE 1:1)



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DREAM

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315b6dd

DRAWING No

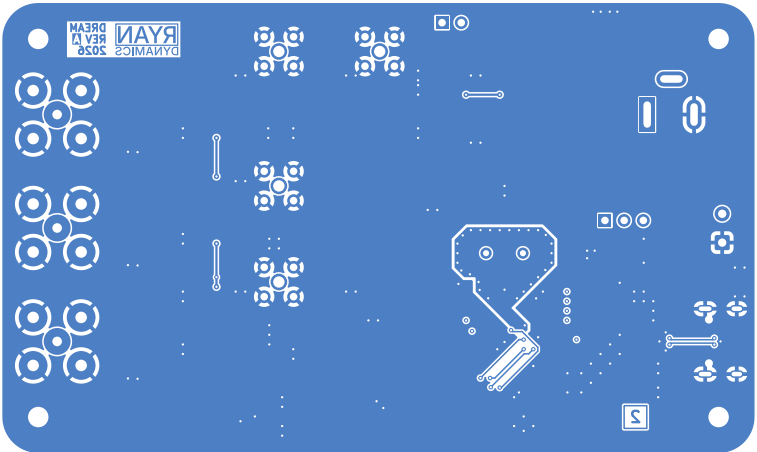
FABRICATION

A3

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DREAM FABRICATION DOCUMENT

L2 (Sig, GND) (SCALE 1:1)



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FILENAME DREAM.kicad_pcb		VARIANT	RELEASED	REVISION 1.2.0	SHEET 8 OF 8	